

Thirawat Bannakulpiphat, M.Eng.

Doctoral Student, Research Unit of Photogrammetry,

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Personal Data

- **Full name:** Thirawat Bannakulpiphat
- **Date of birth:** 7 December 1997
- **Nationality:** Thai
- **Place of birth:** Bangkok
- **Marital status:** Single
- **Family:** I have four siblings including myself, and I am the fourth.
- **Home address:** 81/71 Gusto Village, Soi Thawi Wattana 9, Thawi Watthana, Bangkok, 10170, Thailand

Education

- **Doctoral Programme in Engineering Sciences (Geodesy and Geoinformation)** 2023 – present
Research Unit of Photogrammetry, Department of Geodesy and Geoinformation,
Technische Universität Wien (TU Wien), Vienna, Austria
Advisor: Univ.-Prof. Dipl.-Ing. Dr. tech. Norbert Pfeifer
- **Master of Engineering in Survey Engineering** 2020 – 2022
Faculty of Engineering, Chulalongkorn University, Bangkok, Thailand
Thesis Title: *Modelling of Multi-head Camera Systems on Unmanned Aerial Vehicles for 3D Mapping and Point-Clouds Generation*
GPA: 4.00/4.00
Advisor: Assoc. Prof. Dr.-Ing. Phisan Santitamnont (Leibniz Hannover)
- **Bachelor of Engineering (Second-Class Honors) in Survey Engineering** 2016 – 2020
Faculty of Engineering, Chulalongkorn University, Bangkok, Thailand
Thesis Title: *Analysis of UAV Dual Camera System for 3D Mapping and Building Modeling*
GPA: 3.49/4.00
Advisor: Assoc. Prof. Dr.-Ing. Phisan Santitamnont (Leibniz Hannover)

Areas of Current Research and Interest

Photogrammetry, Computer Vision, UAV Mapping, LiDAR and Laser Scanning, Point-Cloud Processing, Applied Earth Observation, Geoscience and Remote Sensing, Geospatial Programming, and Small-Format Multi-Head Camera Systems

Awards

- Received the Outstanding Graduate Student Award from the Faculty of Engineering and the Department of Survey Engineering (2021).
- Awarded the Royal Thai Government Science and Technology Scholarship (Master's and Ph.D.), allocated by the Ministry of Higher Education, Science, Research and Innovation (2021).
- Winner of the MGA Award in the Rapid Prototype Development (RPD) Challenge 2020–2021, organized by GISTDA and the Cabinet Office, Government of Japan (Topic: *Solution for Disaster Management – Tsunami/Flooding*).
- Selected as a STAR Intern at the Electricity Generating Authority of Thailand (EGAT) Internship 2018 (Topic: *Development of Aerial Photographic Processing Workflow and DEM Generation from Point Clouds*).

Technical Skills

- **Programming Languages:** Python, MATLAB, R
- **Geospatial Software & Tools:** QGIS, ArcGIS, Processing aerial images (PIX4Dmapper, Agisoft Metashape), CloudCompare
- **Databases and Geospatial Libraries:** PostgreSQL, SQLite, SQL, GDAL, PDAL, OpenCV, Open3D, GeoPandas, Rasterio
- **Machine Learning & Deep Learning:** TensorFlow, Keras, Scikit-Learn
- **Data Science & Visualization:** Pandas, Seaborn, Plotly
- **Data Analytics:** Spreadsheets, Data Transformation, Data Visualization, Statistics, Spatial Analysis and Spatial Data Management
- **Web Scraping & Automation:** BeautifulSoup, Selenium
- **Other Tools:** Jupyter Notebook, Google Colab

Past Research and Work Experience

- **Research Assistant - Department of Survey Engineering, Chulalongkorn University**
Bangkok, Thailand
Project: Calibration and Validation of UAV Sensors *Jan 2023 – Sep 2023*
 - Calibrated and validated a small-format multi-head camera system.Project: A Study on Ground Control Targeting for UAV Laser Scanning *Aug 2022 – Dec 2022*
 - Designed and analyzed LiDAR control planes (LCP) for optimization and evaluation.
 - Processed aerial images and point-cloud data from UAV-based laser scanning systems.
- **Research Assistant - Infraplus Co., Ltd** *Aug 2021– Aug 2022*
Bangkok, Thailand
Main Responsibilities
 - Processed UAV aerial images for engineering drawings and developed QA/QC workflows for UAV photogrammetry.
 - Performed post-processing of UAV products supporting road/rail design, superstructure design, environmental impact assessment, and project management.*Additional project:* Measuring, Monitoring, and Evaluating the Construction Measurement for Road and Bridge Works in the Sloping Area Using SAR Interferometry and Hyperspectral Data
 - Processed satellite imagery for monitoring slope, road, and bridge movement using time-series InSAR techniques (SBAS and PSI).
 - Processed UAV imagery acquired with multispectral camera systems.
- **Research Assistant - Chula Unisearch, Chulalongkorn University** *Aug 2020 – Jul 2021*
Bangkok, Thailand
Project: Design and Development of UAS VTOL with a Dual-camera for Wide-Area HV Transmission Line Survey
 - Processed aerial images from UAV multi-head camera systems.
 - Calibrated and validated the rigged camera model.
 - Analyzed the camera-rig model and GNSS antenna configuration for camera offset determination.
- **Research Assistant - Water Resource System Research Unit, Chulalongkorn University** *May 2020 – Nov 2020*
Bangkok, Thailand
Project: Open Data Remote Sensing Development Study for Water Resources Management Research
 - Processed satellite imagery using Python to extract water boundaries.
 - Delivered demonstrations and training sessions on using the operating system to download satellite images.

- **Engineering Trainee - Electricity Generating Authority of Thailand (EGAT)** *May 2019 – Jul 2019*
Bangkok, Thailand
Survey Division, Geoinformatics Department, Photogrammetry and Remote Sensing Section
 - Processed UAV aerial images for engineering workflows and applied production engineering methods.
 - Prepared a user manual for Agisoft PhotoScan.

Teaching

- **Guest Lecturer - Department of Survey Engineering, Chulalongkorn University** *Feb 2023 – Sep 2023*
Bangkok, Thailand
 - 2108202 Surveying Engineering (Taught field practices)
 - 2108326 Geospatial Information Analysis and Visualization (Taught data-processing practices)
 - 2108373 Advanced Photogrammetry (Taught theoretical concepts and processing workflows)
 - 2108421 Modern Integrated Surveying Technology (Taught theoretical and practical components)
 - 2108412 Satellite Surveying (Taught field practices)
- **Co-Lecturer - Free and Open-Source Software for Geospatial (FOSS4G) Thailand 2022** *2022*
26 November 2022, Khon Kaen, Thailand
 - Co-instructed with Assoc. Prof. Dr.-Ing. Phisan Santitamnont on point-cloud processing using FOSS and Python tools.
- **Teaching Assistant (TA) - Department of Survey Engineering, Chulalongkorn University** *2020 – 2021*
Bangkok, Thailand
 - 2108301 Surveying with Construction Applications (Co-instructed lectures and field practices)
 - 2108373 Advanced Photogrammetry (Assisted in teaching aerial image processing practices)

Extracurricular Activities

- **Organizing committee - 3rd International Workshop “3D Underwater Mapping from Above and Below” 2025**
8 - 11 July 2025, Vienna, Austria
 - Served as part of the on-site organizing team, as listed in the official event program.
 - Supported coordination and logistics for tutorials, technical sessions, and overall workshop operations.
- **Guest Speaker - Academic Seminar & Workshop “Thai Researchers’ Life in Central & Eastern Europe”** *2025*
5 July 2025, Budapest, Hungary
 - Invited as a guest speaker at the first academic seminar and workshop organized by the Thai Student Association in Hungary in collaboration with the Royal Thai Embassy in Budapest.
 - Presented research on UAV-based 3D mapping and participated in a panel discussion with researchers from multiple disciplines across Central and Eastern Europe.
 - Contributed to a hands-on academic writing workshop, supporting participants in improving scientific communication and research presentation skills.
- **President - THai Students and Researchers Club in Austria (THSRCA)** *2024 – 2025*
Vienna, Austria
 - Developed a comprehensive handbook to guide Thai students and researchers through key administrative procedures before and during their studies in Austria.
 - Organized community-building activities that helped connect Thai students and researchers across Austria.
 - Co-founded the Thai Students Association in Europe (TSAE), promoting collaboration and cultural exchange among Thai students, scholars, and researchers across Europe.

Publications

Google Scholar: <https://scholar.google.com/citations?user=ZZ1vnTYAAAAJ&hl=en&authuser=1>

Journal Article and Conference Paper

1. **Bannakulpiphat, T.**, Karel, W., Ressler, C., Pfeifer, N. (2025). Evaluation of Bundle Block Adjustment for 3D Geospatial Modelling Using UAV-Based Small-Format Multi-Head Camera Systems, *RevCAD Journal of Geodesy and Cadastre*, 39, 13–24.
2. **Bannakulpiphat, T.**, Karel, W., Ressler, C., and Pfeifer, N. (2024). Optimizing Bundle Block Adjustment for High-Overlap Small-Format Multi-Head Camera Systems, *Int. Arch. Photogramm. Remote Sens. Spatial Inf. Sci.*, XLVIII-2/W7-2024, 17–24.
3. **Bannakulpiphat, T.**, and Santitamnont, P. (2024). A Comparative Analysis of Camera Rig Parameters in Photogrammetric Software for Small-Format Oblique Camera System on Unmanned Aerial Vehicle. *International Journal of Geoinformatics*, 20(2), 1–10.
4. **Bannakulpiphat, T.**, Santitamnont, P., Petchrungrussamee, P., Srirattanapaisarn, T., Puatanachokchai, C., and Santitamnont, P. (2023). Design of Gable Roof LCP for Geometric Evaluation of Point-cloud. *Engineering Journal of Research and Development*, Volume 34 Issue 4.
5. **Bannakulpiphat, T.**, and Santitamnont, P. (2023). Enhancing Performance and Evaluating Accuracy of GNSS Positioning for UAV Mapping. *Engineering Journal of Research and Development*, Volume 34 Issue 3.
6. **Bannakulpiphat, T.**, and Santitamnont, P. (2023). New Era of Mapping Products from UAV-Based Oblique Camera System. *The 5th International Conference on Civil and Building Engineering Informatics*, Bangkok, July 2023, pp. 231-236.
7. **Bannakulpiphat, T.**, Santitamnont, P., Maneenart, T., and Wongweeranimit, W. (2023). Best Practice for Mapping Production from UAV Imagery. *Engineering Journal of Research and Development*, Volume 34 Issue 1.
8. **Bannakulpiphat, T.**, Santitamnont, P., and Maneenart, T. (2022). A Case Study of Multi-head Camera Systems on UAV for The Generation of High-Quality 3D mapping. *Engineering Journal of Research and Development*, Volume 33 Issue 4.
9. Santitamnont, P., **Bannakulpiphat, T.**, and Junlanak, B. (2022). Design Principles for Low Distortion Map Projection over a Project. *Engineering Journal of Research and Development*, Volume 33 Issue 4.
10. Kimtan, P., **Bannakulpiphat, T.**, Boonrasi, P., and Khoonpool, R. (2023). A Study of The Depth of The Channel Using A Low-cost GNSS Receiver Built into A Fish-Finder Sounder: A Case Study of Songkhla Lake, Songkhla Province. *The 28th National Convention on Civil Engineering*.
11. Srirattanapaisarn, T., Santitamnont, P., and **Bannakulpiphat, T.** (2023). A Comparative Study of The Point-cloud Quality from UAV Laser Scanning and Block-based Images Processing. *Engineering Journal of Research and Development*, Volume 34 Issue 2.
12. Srinoi, T., **Bannakulpiphat, T.**, and Santitamnont, P. (2024). Building Height Estimation Production from Open Satellite Imagery by Gradient Boosting Regression Technique. *Engineering Journal of Research and Development*, Volume 35 Issue 2.
13. Srinoi, T., **Bannakulpiphat, T.**, and Santitamnont, P. (2024). Building Footprint Extraction from True Orthophoto by Geospatial Segment Anything Model. *Engineering Journal of Research and Development*, Volume 35 Issue 2.
14. Srinoi, T., **Bannakulpiphat, T.**, and Santitamnont, P. (2024). Satellite Derived Bathymetry Production Case Study at Deepwater Port of Maptaphut Industrial Park Rayong. *Engineering Journal of Research and Development*, Volume 35 Issue 1.
15. Srinoi, T., **Bannakulpiphat, T.**, Santitamnont, P., and Vaiphasa, C. (2024). Building Height Estimation from Open Optical Remote Sensing by Machine Learning Regression Technique: A Case Study of the Central of Bangkok. *International Journal of Geoinformatics*, 20(9), 43–53.

Magazines

1. **Bannakulpiphat, T.** (2025). UAV-Based 3D Mapping with Small-Format Multi-Head Camera Systems. *Journal of Thai Student's Association in Europe*, Volume 1 Issue 1.
2. **Bannakulpiphat, T.** (2020). Analysis of UAV Dual Camera System for 3D Mapping and Building Modeling. *Intania Magazine of Chulalongkorn University*, No.2/2020

Additional Information

- **Preferred name:** Thirawat Bannakulpihat, Tae Bannakulpihat, Tae
- **Telephone:** (+66) 80 693 6553 [Thailand], +43 677 616 34773 [Austria]
- **LinkedIn:** <https://www.linkedin.com/in/thirawat-bannakulpihat-318a9719b/>
- **GitHub:** <https://github.com/ThirawatBan>
- **Medium:** <https://medium.com/@thirawatbannakulpihat>